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Python Programming Concepts

This presentation covers fundamental concepts in Python programming, including keywords, input/output operations, mathematical functions, and control flow structures like loops.

Python Keywords

All keywords in Python:

Keywords

False, None, True, and, as, assert, async, await, break, class, continue, def, del, elif, else, except, finally, for, from, global, if, import, in, is, lambda, nonlocal, not, or, pass, raise, return, try, while, with, yield

These keywords are reserved words in Python and cannot be used as identifiers (variable names, function names, etc.). They have special meanings and are used to define the syntax and structure of the Python language.

Input and Calculations in Python

Finding the Largest of Three Numbers

wap in python to to input three numbers and find the largest

```
num1 = float(input("Enter first number: "))
num2 = float(input("Enter second number: "))
num3 = float(input("Enter third number: "))
if (num1 >= num2) and (num1 >= num3):
    largest = num1
elif (num2 >= num1) and (num2 >= num3):
    largest = num2
else:
    largest = num3
print("The largest number is", largest)
```

Calculating Factorial of an Integer

awp in python to input integers and find factorial

```
num = int(input("Enter a non-negative integer: "))
factorial = 1
if num < 0:
    print("Factorial does not exist for negative numbers")
elif num == 0:
    print("The factorial of 0 is 1")
else:
    for i in range(1, num + 1):
        factorial = factorial * i
    print("The factorial of", num, "is", factorial)
```

These examples demonstrate how to take user input and perform basic calculations like finding the maximum value or computing a factorial using conditional statements and loops.

Understanding Loops in Python

define loop mention different types of loops with examples in python

A loop is a programming construct that allows a block of code to be executed repeatedly based on a certain condition or for a specific number of times. Loops are essential for automating repetitive tasks and iterating over collections of data.

1. For Loop

Used for iterating over a sequence (like a list, tuple, dictionary, set, or string) or other iterable objects. It executes a block of code for each item in the sequence.

```
fruits = ["apple", "banana", "cherry"]
for x in fruits:
    print(x)
```

2. While Loop

Used to execute a block of code repeatedly as long as a given condition is true. The loop continues until the condition becomes false.

```
i = 1
while i < 6:
    print(i)
    i += 1
```

Loops are fundamental control flow statements that enable efficient and concise code for repetitive operations.